SCANNER BRAIN ACTIVATION DURING MENTAL STRESS AND AMBULATORY BLOOD PRESSURE RESPONSES TO STRESS DURING DAILY LIFE

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Exaggerated cardiovascular reactivity (CVR) to mental stress is associated with increased risk for hypertension and atherosclerosis. Using a machine learning approach in a prior sample of adults who completed a standardized fMRI stressor battery, Gianaros et al. (2017) cross-validated a multivariate brain pattern that predicted exaggerated CVR in-scanner. In the current study, we used the same fMRI stressor battery to measure the expression of this brain pattern in a new sample of 279 healthy midlife adults from the Neurobiology of Health (NOAH) study (ages 28-56). In preregistered analyses, we addressed the question of whether this brain pattern could identify those most susceptible to exaggerated ambulatory blood pressure responses during mental stress in daily life.

We used data from 2 sets of procedures to address this question. First, participants completed 4 days of ambulatory blood pressure (ABP) monitoring during which we collected hourly ABPs along with smartphone-based diary assessments measuring daily life “task strain” events (high demand, low control) along with demographic (age, sex, race, education) and time- varying covariates (posture, activity, substance use) (Kamarck et al., 2018). Second, participants completed an fMRI stressor battery from which we extracted a dot product score representing the extent to which participants expressed the previously validated fMRI brain pattern trained to predict exaggerated CVR in the laboratory (Cardiovascular Reactivity to Stress Pattern expression or CRSP-e score). In multilevel models, we examined whether the magnitude of each person’s systolic BP response to task strain events during daily life was moderated by their CRSP-e score, after covariate adjustment.

The effect of task strain on momentary SBP was, indeed, significantly moderated by CRSP-e scores (b for strain by CRSP-e interaction = .39, F (1, 14,000) = 4.0, P = .046). There was a dose-response effect, with the association between task strain and SBP increasing with successively larger CRSP-e scores (For CRSP-e 2 sd below the mean, strain b = .47, P = .7; for -1 sd, strain = 1.03, P = .009; for + 1 sd, strain b = 2.15, P < .0001; for +2 sd, strain b = 2.70, P < .0001). There was also significant shared variance between fMRI CRSP-e scores and SBP reactivity to task strain in daily life (r = .35, n = 279, P < .0001). This is the first evidence that individual differences in stress-related brain patterns measured by fMRI generalize to predict ambulatory CVR to mental stress in daily life. Ongoing work is examining this stress-related brain activation pattern as a marker of stress sensitivity and disease risk. This protocol was preregistered in OSF. Supported by P01HL040962.

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ANGER ALLOSTASIS IN EVERYDAY LIFE: TESTING INERTIA IN MOMENTARY ANGER AND AMBULATORY BLOOD PRESSURE ASSOCIATIONS AS FUNCTION OF TRAIT ANGER

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Background: Having a predisposition to anger is a risk factor for cardiovascular disease and mortality, in part due to the known associations between experiencing anger and elevated blood pressure (BP). Yet, the risk from trait anger may run deeper than just a tendency to experience anger. Drawing from the theory of allostatic load, this paper proposes an anger allostasis model, positing that a core feature of trait anger that poses risk is an unresponsiveness (or inertia) to one’s context.

Purpose: The present study examined concurrent and cross-lagged associations between momentary anger and ambulatory BP, testing if trait anger moderates these associations in a way that suggests allostasis: inertia in moment-to-moment anger and BP patterns, and reduced reactivity and recovery across anger to BP associations.

Method: The sample comprised 257 participants recruited from the New York City area (53% female; age M = 41 y; 56% Black). At baseline, participants completed a measure of trait anger. Following, participants wore a SpaceLabs ambulatory BP monitor for 24 hours that collected readings every 30 minutes during waking hours (~20 assessments per person). Along with each BP reading was an ecological momentary assessment of one’s current anger state. A two-level dynamic structural equation model was run to examine possible allostasis: anger and/or ambulatory BP inertia (e.g., anger at time t-1 predicting anger at time t), anger and ambulatory BP cross-lags (e.g., anger at time t-1 predicting BP at time t), and anger and ambulatory BP cross associations (i.e., anger and BP at time t). Trait anger and age were entered as moderators of these associations.

Results: Suggesting anger allostasis, as trait anger increased the following combination emerged: participants reported more overall state anger (b = .40, SD = .20, P = .023), reduced time t BP reactivity to time t anger (cross associations: b = -.01, SD = .003, P = .007); and reduced recovery from time t-1 BP to time t anger (cross-lagged

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